





ORIGINAL ARTICLE

Energy Absorption of 3D Printed ABS and TPU Multimaterial Honeycomb Structures

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Abstract

Advances in multimaterial 3D printing are enabling the construction of advantageous engineering structures that benefit from material synergies. Cellular structures, such as honeycombs, provide high-energy absorption to weight ratios that could benefit from multimaterial strategies to improve the safety and performance of engineered systems. In this study, we investigate the energy absorption for honeycombs with square and hexagonal unit cells constructed from acrylonitrile butadiene styrene (ABS) and thermoplastic polyurethane (TPU). Honeycombs were fabricated and tested for out-of-plane and in-plane compression using ABS, TPU, and a combination of ABS with a central TPU band of tunable height. Out-of-plane energy absorption for square honeycombs increased from 2.2 kN·mm for TPU samples to 11.5 kN·mm for ABS samples and energy absorption of hexagonal honeycombs increased from 2.9 to 15.1 kN·mm as proportions of TPU/ABS were altered. In-plane loading demonstrated a sequential collapse of unit cell rows in square honeycombs with energy absorption of 0.1 to 2.6 kN·mm and a gradual failure of hexagonal honeycombs with energy absorption of 0.6 to 2.0 kN·mm. These results demonstrate how multimaterial combinations affect honeycomb compressive response by highlighting their benefits for controlled energy absorption and deformation for tunable performance in diverse engineering applications.

Keywords: 3D printing, fused deposition modeling, multimaterial printing, design, honeycomb, mechanics

Introduction

THE ADVENT OF ADDITIVE MANUFACTURING, also known as three-dimensional (3D) printing, has opened the door for customizing mechanical designs using a diverse range of structures and materials.^{1,2} The mechanical performance of 3D-printed parts depends on their constituent materials, printing process, and designed structure.³ Deposition of multiple materials throughout a designed part is an emerging strategy that enables synergistic use of contrasting materials to improve overall system performance.^{4–6} Multimaterial fabrication is possible with fused deposition modeling (FDM) processes that produce parts with directed polymer deposition of one material at a time that fuses with another.⁷ FDM is advantageous for its low cost and ease of use compared with other multimaterial printing technologies. Com-

mon 3D printing materials for FDM include acrylonitrile butadiene styrene (ABS) for its high stiffness and thermoplastic polyurethane (TPU) for its hyperelasticity.^{8–10}

In this study, we investigate multimaterial combinations of ABS and TPU in mechanically efficient honeycomb cellular structures to provide a tailored mechanical response suitable for biomedical applications, such as helmets and prosthetics.^{11,12} The investigation is supported by mechanical testing experiments to measure the response of multimaterial honeycombs with varied designs and material combination strategies.

Honeycombs are a type of cellular lattice structure that enables high energy absorption^{13,14} and have applications for improving shock absorption.¹⁵ Honeycombs are a prismatic extrusion of a two-dimensional shape from repeating unit cells that form larger structures with effective mechanical

properties that depend on their unit cell topology.^{16–18} Lattice systems are often found naturally due to their mechanical efficiency making them desirable to mimic for engineered systems.^{19–21} For man-made structures, truss-based lattices have high stiffness,²² while foams and honeycombs provide capabilities for energy absorption.^{23,24} Determining the mechanical properties of 3D-printed lattice structures is facilitated by the design and testing of lattices.^{25–29} Lattices are widely used because of their amenability for tuning mechanical systems.³⁰ For instance, multimaterial helmets or prosthetics with tailored mechanics are customizable on a per-print basis for a user's specific physiology.^{31,32}

These multimaterial systems provide a contrast to density-graded systems where the structure changes throughout a design to enable multiple functionalities for lightweight structures.³³ In this study, we propose a novel combination of elastic and stiff materials to provide tunable functionalities throughout structures such as prosthetics (Fig. 1), where design tailoring commences through altering the relative proportions of each material.

In these scenarios, ABS could provide high ultimate strength with stiffness, while a hyperelastic TPU material could provide large deformation before failure, thereby enabling a preferred overall system response. Varying the amounts of these contrasting materials provides an opportunity to customize designs based on the weight/size of the person or to reduce jerk transfer to the body with efficient energy absorption.¹¹ The honeycombs with square and hexagonal unit cells in Figure 1 are desirable structures due to their simplicity and highly efficient design,³⁴ therefore mitigating the effects the 3D printing process may incur on measured design performance due to printing defects in more complex structures.^{25,26} Honeycombs generally have mechanical efficiency for high energy absorption based on their relative density,^{30,31} which makes them well suited for impact biomechanical applications.

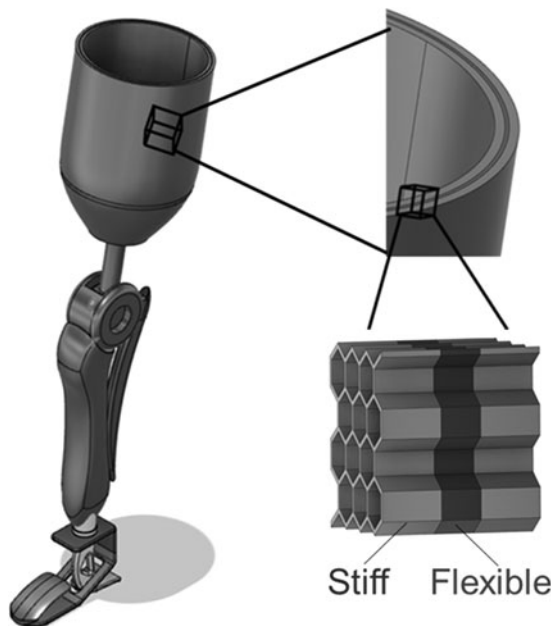


FIG. 1. Representative prosthetic leg application with embedded multimaterial honeycombs constructed from stiff and flexible materials.

The energy absorption behavior of 3D-printed honeycombs is dictated by their material, cell size, structure size, and wall thickness.^{13,35} Honeycomb design parameters that have empirically resulted in mechanical property differences include cell length and internal angle.³⁴ The repetitive energy absorption and recovery of honeycombs printed with TPU material have shown a high-specific energy absorption with tunable stiffness based on their wall thickness.³⁶ An advantage of adding a second material to form a multimaterial structure is the possibility of altering a design's performance without the need for change in its volume.³¹ The application of a multimaterial approach could also help in vibration isolation in the flexible material while retaining stiffness otherwise.³⁷ However, before designing for a tuned response, there is a need to study the material mechanics to quantify their properties to inform design decisions, which is still poorly understood for applications with FDM.

The failure mechanics for honeycombs for out-of-plane and in-plane compression has been studied extensively.³⁸ However, the introduction of 3D printing can cause designs to perform differently based on printing parameters and part orientation.³⁹ Due to these factors, there is a need to investigate how much energy is absorbed with differently printed parts before failure since they deviate from a theoretically ideal structure. A standard approach for measuring energy absorption in honeycombs is calculating the area under the force–displacement curve before densification.³⁴ Since alterations in a design will change energy absorption behavior, and real-world cases such as crashes often have uncontrolled loads, it is important to test lattice in multiple orientations.⁴⁰ Commonly, honeycombs are tested to characterize properties out-of-plane and in-plane, which is determined based on the orientation of a load relative to the axis of layered unit cells.

The goal of this article is to investigate, experimentally, the compressive behavior and energy absorption of multimaterial 3D-printed honeycombs with different layouts of stiff and flexible materials, namely ABS and TPU. The materials are arranged in layouts that vary the band thickness of TPU sandwiched between ABS materials. Both out-of-plane and in-plane compression tests are performed with energy absorption analysis. Imaging during compression is conducted to determine how different layouts deform during compression, which demonstrates behaviors of lattices relevant to engineering applications. As a whole, the study seeks to provide fundamental mechanics knowledge for how multimaterial lattices behave that enables tailoring of preferred properties in engineered systems.

Materials and Methods

Honeycomb design

Lattices were designed with honeycombs with square or hexagon unit cells (Fig. 2). The computer aided design model was designed in Fusion 360 (Autodesk Inc., USA). Cura software by Ultimaker (Ultimaker, Netherlands) was used to assign multiple materials throughout the lattice to form bands across its cross section. Throughout the study, the lattice dimensions are retained, while the material layout is varied by altering the TPU band height H_b . Figure 2 shows the design parameters for configuring square and hexagonal honeycombs.

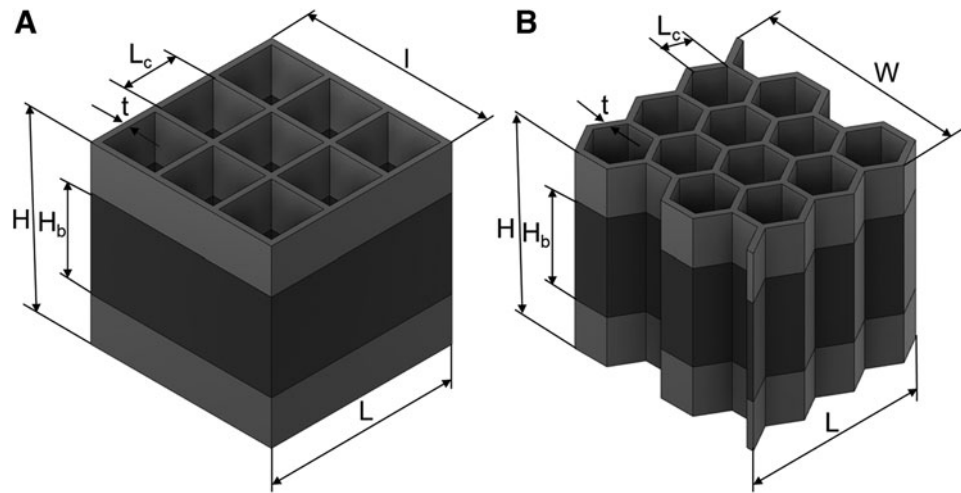


FIG. 2. Design parameters. (A) *Square* honeycomb. (B) *Hexagonal* honeycomb; gray material indicates the placement of ABS and blue material indicates the placement of TPU. ABS, acrylonitrile butadiene styrene; TPU, thermoplastic polyurethane.

For both structures, the wall thickness t was specified as 0.4 mm. A patterning of three-unit cells in each lateral direction for square honeycombs with a unit cell length L_c of 3.6 mm is used. The lattice Length L , Width W , and Height H were 12.4 mm by 12.4 mm by 12 mm, respectively. Hexagonal honeycombs were generated with similar design parameter values and overall dimensions by patterning unit cells laterally, with all parameter values provided in Table 1. Both honeycombs are designed within 5% weight difference to control for how material volume may affect mechanics. The TPU band height was altered from 0 mm to 12 mm in increments of 3 mm to investigate how different proportions of each material affect the bulk structural response.

Printing process

All samples were printed using FDM with an Ultimaker S3 multimaterial printing system using multiple nozzles and materials. The samples were printed with Ultimaker ABS black (Ultimaker) and TPU 95A blue (Ultimaker) using a 2.6-mm-diameter filament and a 0.4 mm AA type nozzle. The layer thickness was 0.2 mm with a printing speed of 55 mm/s for ABS and 25 mm/s for TPU, with 100% flow rate. The bed temper-

ature was kept at 85°C with adhesive glue spread on the printing bed for improved bed adhesion. The solid samples were printed with 100% infill density with line-type infill with four peripheral walls. Printing parameter values are summarized in Table 2.

Figure 3 demonstrates the printing process for building each structure in relation to the printhead, axial coordinates, and build direction of the sample b_z . The XY plane is considered the in-plane direction, and the z -direction the out-of-plane direction that is aligned with the layer-by-layer stacking of material to create the structure aligned with the z -axis.

A priming tower is also printed; the main purpose of the priming tower is to reduce blob formation and ensure that the printing material is ready for each layer. After printing the samples, an optical microscope (Olympus DSX 510) with 5× zoom was used for imaging.

Mechanical testing

The mechanical response of samples was studied using the uniaxial compression test with a constant velocity of 100 mm/min, which is at the strain rate of 0.13 s^{-1} .⁴¹ The size of the honeycombs is in agreement with ASTM D1621,⁴² with the compression test stopping at 9.6 mm displacement,

TABLE 1. DESIGN PARAMETERS AND RESPECTIVE VALUES

Design parameter	Unit cell type	Value (mm)
Unit cell length (L_c)	Square	3.6
	Hexagon	3.1
Wall thickness (t)	Square	0.4
	Hexagon	0.4
Lattice length (L)	Square	12.4
	Hexagon	13.8
Lattice width (W)	Square	12.5
	Hexagon	13.5
Lattice height (H)	Square	12.0
	Hexagon	12.0
Band height (H_b)	Square	0–12
	Hexagon	0–12

TABLE 2. PRINTING PARAMETERS AND RESPECTIVE VALUES

Printing parameter	Material	Value
Nozzle diameter	ABS/TPU	0.4 mm
	ABS	240°C
Nozzle temperature	TPU	223°C
	ABS/TPU	80°C
Build plate temperature	ABS/TPU	0.2 mm
Layer height	ABS/TPU	100%
Print infill	ABS/TPU	Lines
Infill type	ABS/TPU	Enabled
Retraction	ABS/TPU	55 mm/s
Printing speed	ABS	25 mm/s
	TPU	

ABS, acrylonitrile butadiene styrene; TPU, thermoplastic polyurethane.

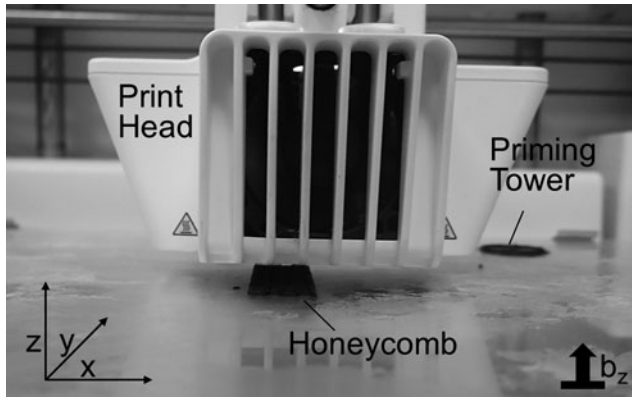


FIG. 3. Printing process demonstrating orientation of honeycomb relevant to the printhead.

which is 80% of the original height. An Instron 5966 Universal testing machine (Instron Engineering Corporation, USA) with a load cell of 10 kN was used to conduct the testing. The data for displacement versus the load curve were extracted and analyzed for energy absorption using custom MATLAB code.³⁴ Testing was conducted for three printed samples for each design configuration and orientation. The out-of-plane testing direction is defined as the load applied normal with the z -axis of the print during the printing process (Fig. 4A) and in-plane as the load applied normal to the y -axis (Fig. 4B).⁴³

Lattice mechanics depends on combinations of ABS and TPU material due to their contrasting properties, as summarized in Table 3, provided by the material manufacturers. The table demonstrates the different properties of ABS as a linear elastic material and TPU as a hyperelastic material that will lead to differing displacement and energy absorption responses as their relative proportions in a material are altered.^{44,45}

The energy absorbed (E) by samples was calculated as the area under the force–displacement curves using the following formula³⁴:

TABLE 3. MECHANICAL PROPERTIES OF PRINTING MATERIAL FROM TECHNICAL DATASHEET

	ABS (XY flat)	TPU (XY flat)
Young's modulus (MPa)	1962 ± 31	67 ± 6
Yield strength (MPa)	38.1 ± 0.3	No yield
Elongation at break (%)	4.6 ± 0.3	>560

$$E = \int_0^{\delta} F d\delta \quad (1)$$

where E is the energy absorbed under the load for compressive displacement δ while F is the applied load. Energy absorption was calculated for honeycomb lattices where displacement was fixed at 9.6 mm for a strain of 0.8 for all samples tested. All data were reported as the mean of three samples tested with error bars representative of the standard deviation. A pairwise t -test is performed to compare the means between samples to determine significance when $p \leq 0.05$.

Results

Fabrication quality

Initially, printing quality was investigated since FDM can cause artifacts that influence mechanics, such as introducing voids due to the deposition of material with a circular nozzle.^{46,47} Printed samples are demonstrated in Figure 5 for square and hexagonal honeycombs. All prints were qualitatively accurate, with some obvious defects such as extra stringing material that generally is nonstructural compared with microscopic voids that could reduce stiffness. Microscopic defects are observed in Figure 6 that highlights the presence of voids and stringing.

In Figure 6, the microscopic voids in walls are potentially caused by underextrusion or melting of previous deposited layers. Hexagonal samples tended to have more stringing

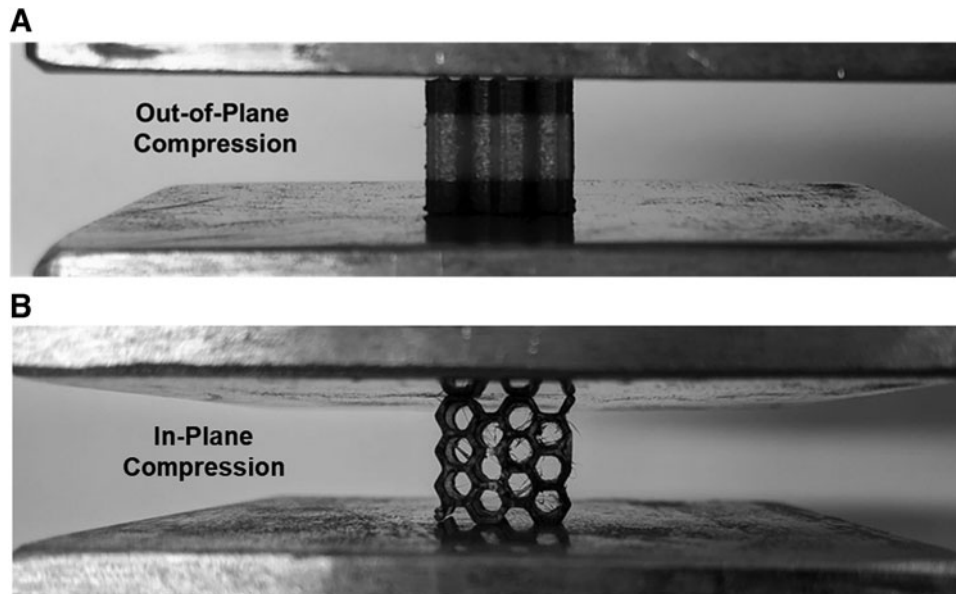


FIG. 4. Lattices configured between compression plates for (A) out-of-plane and (B) in-plane testing orientations.

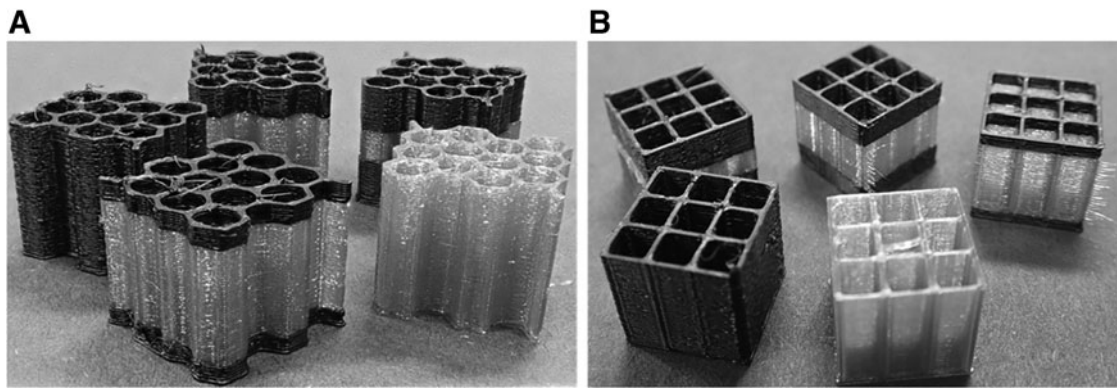


FIG. 5. Three-dimensional-printed samples for (A) *square* honeycombs with varied TPU thicknesses and (B) *hexagonal* honeycombs with varied TPU band thicknesses.

defects compared with square honeycombs possibly due to the nozzle having to follow a more complex geometry each layer. Overall, the print quality of samples retained the intended structure, and defects were generally similar for square/hexagon lattices, suggesting that any difference in their mechanics is primarily due to their design.

Out-of-plane response

The mechanics of all samples with pure and mixed materials for square and hexagonal honeycombs was tested for out-of-plane response, with square honeycomb results demonstrated in Figure 7. Figure 7A illustrates the failure of an ABS-only sample, Figure 7B for a multimaterial sample, and Figure 7C for a purely TPU sample. Results demonstrate wall buckling, with the TPU deforming to a greater extent than the ABS in the multimaterial sample in Figure 7B.

These behaviors lead to the differing peak loads occurring at higher displacements as TPU band size is increased in Figure 7D. The ABS-only sample has the highest stiffness

and load-carrying capacity before the first peak load, while the TPU-only sample has the lowest. Although a clear yield point is not defined for multimaterial samples because of the nonlinear stress–strain behavior of TPU,⁴⁵ increasing the thickness of the TPU band decreases the stiffness of the sample while moving the first peak to a higher displacement. This occurs because the TPU compresses and displaces initially until a point where the ABS compresses and exhibits yielding for that portion of the structure causing the second peak before the densification. However, as TPU thickness increases to a nearly full TPU sample, the mechanical response tends to become more similar to the TPU-only samples with no identifiable peak load/yield stress. These different qualities of the load–displacement curve affect the average energy absorption of each sample plotted in Figure 7E.

The energy absorption was highest for the 0 mm TPU (ABS-only) samples with 1.15 ± 1.02 kN·mm energy absorbed and decreased as the TPU band thickness is increased until reading the lowest energy absorption of 2.18 ± 0.35 kN·mm for 12 mm

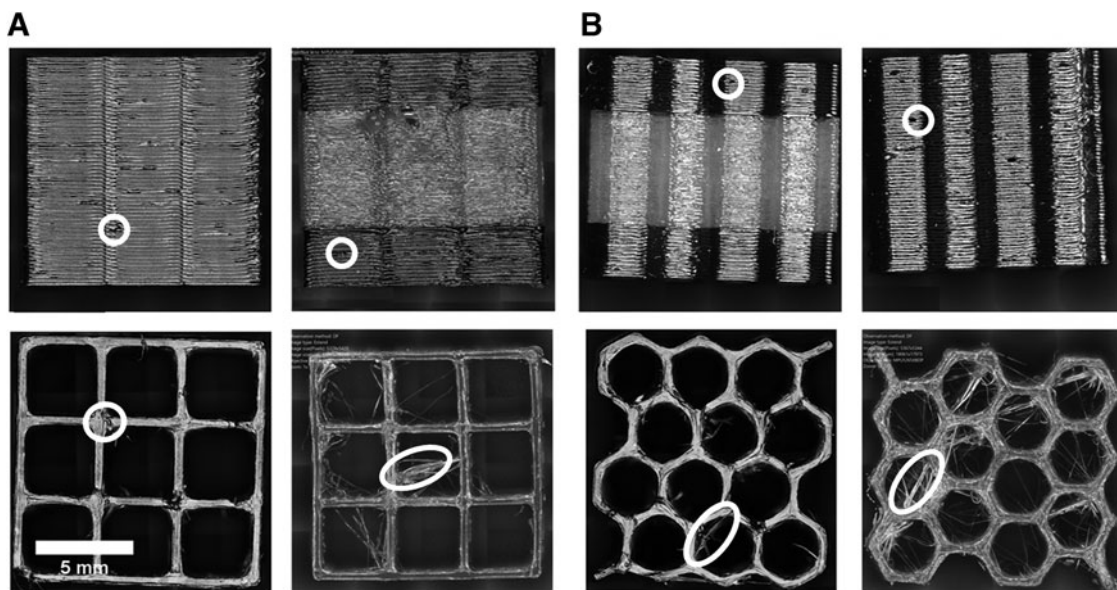


FIG. 6. Microscopy of printed parts for (A) *square* honeycombs of pure ABS and ABS/TPU and (B) *hexagonal* honeycombs of ABS/TPU and pure TPU materials from *side* and *top* perspectives. Exemplary print defects are highlighted with *red circles*. Scale bar ~ 5 mm.

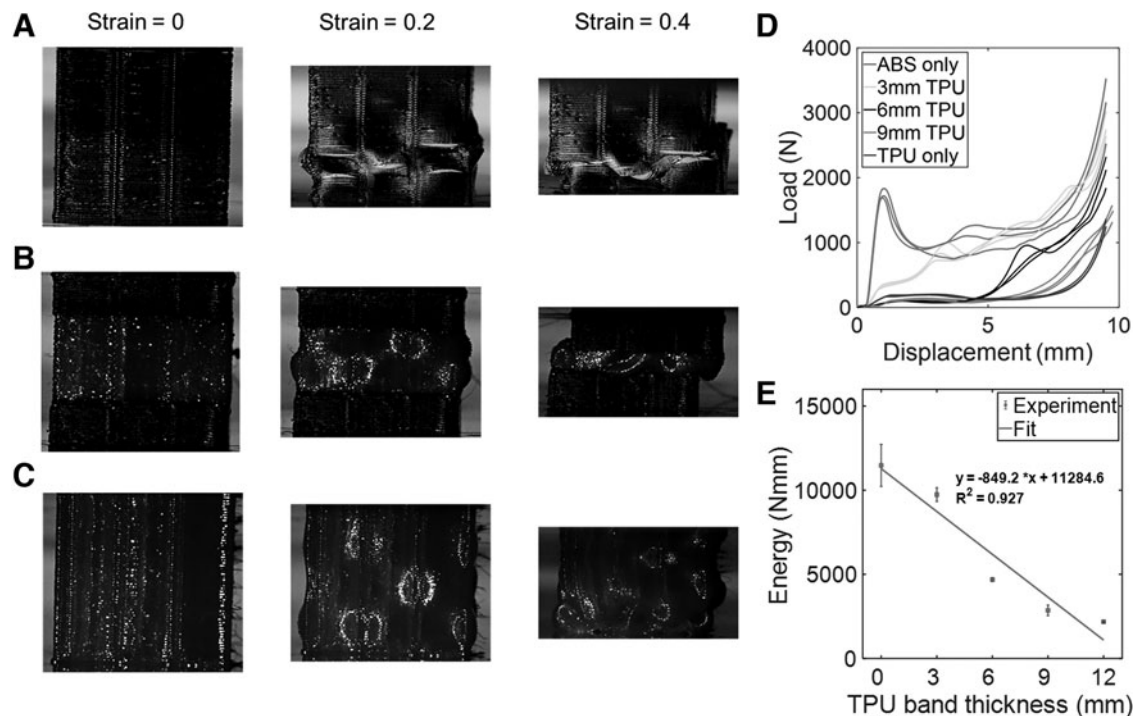


FIG. 7. Out-of-plane compression for *square* honeycombs for a (A) pure ABS honeycomb, (B) 50% ABS and 50% TPU multimaterial honeycomb, and (C) pure TPU honeycomb. (D) Load–displacement response of honeycombs with (E) energy absorption.

TPU (TPU-only) samples. When averaging these extremes, the expected middle point is about $6 \text{ kN} \cdot \text{mm}$, however, the sample that is half ABS and half TPU by material proportions has a lower energy absorption of $4 \text{ kN} \cdot \text{mm}$, demonstrating that the behaviors do not obey a simple law of averages. A linear curve fitting based on the experimental data is also shown in Figure 7E, with the coefficient of determination (R^2) value of 0.927 that suggests a linear trend is predictive of the behavior generally, but adjustments are necessary to improve modeling accuracy.

The mechanical response of hexagonal samples is shown in Figure 8 for all designs tested. It is observed that the deformation of ABS (Fig. 8A) is similar to a square honeycomb, the TPU deforms first in the multimaterial samples (Fig. 8B), while the TPU-only samples failed by buckling under the load (Fig. 8C). Figure 8D shows the load–displacement behavior of different hexagonal designs, with ABS-only samples having a distinctly higher stiffness and compressive strength. In addition, a similar trend as the square honeycomb design is seen in energy absorption for the hexagonal honeycomb, with the highest energy absorption for ABS-only sample of $15.11 \pm 0.48 \text{ kN} \cdot \text{mm}$ energy absorption at 0.8 strain and lowest for TPU-only sample of $2.91 \pm 0.12 \text{ kN} \cdot \text{mm}$ (Fig. 8E).

There are some differences between the square and hexagonal honeycombs in terms of peak load–displacement and energy absorbed. In all material cases, the hexagonal honeycombs have a significantly higher energy absorption ($p \leq 0.05$) than the square honeycomb, except for the 3 mm TPU band case, where there was no significant difference. In all other cases, the hexagon had 30–66% higher energy absorption than the square topology designed with the same

materials. These results suggest that a tuned load–displacement response can be achieved by changing the TPU band thickness in the structure and the hexagon shape is preferred to the square unit cell to improve overall response.

In-plane response

The in-plane mechanical response was also investigated for the structures of the same designs for out-of-plane testing to understand how the multibehavior structures behave in a second orientation. It was observed that in-plane failure for square honeycomb samples occurred with three load peaks that correspond to each layer of unit cells failing sequentially (Fig. 9A). These sequential failures represent the spikes on the load–displacement curve due to the buckling of the wall for each layer (Fig. 9B). The bottom layer bears the initial load until the first spike when the wall buckles and the load falls, then the top layer wall bears the load until the second spike when the wall buckles and the load decreases, followed by a similar pattern that happens to the middle layer causing three spikes. The in-plane energy absorption was calculated for each design and is presented in Figure 9C.

The energy absorption for in-plane testing follows a similar trend as the out-of-plane compression test, with maximum energy absorbed by the ABS-only sample ($2.58 \pm 0.32 \text{ kN} \cdot \text{mm}$) and lowest for TPU-only samples ($0.12 \pm 0.018 \text{ kN} \cdot \text{mm}$). The energy absorption behavior of the structures for in-plane compression is fitted linearly with an R^2 value of 0.928.

The in-plane compression of the hexagonal honeycomb with different TPU band thicknesses was studied next (Fig. 10). Figure 10A shows the deformation behavior for the

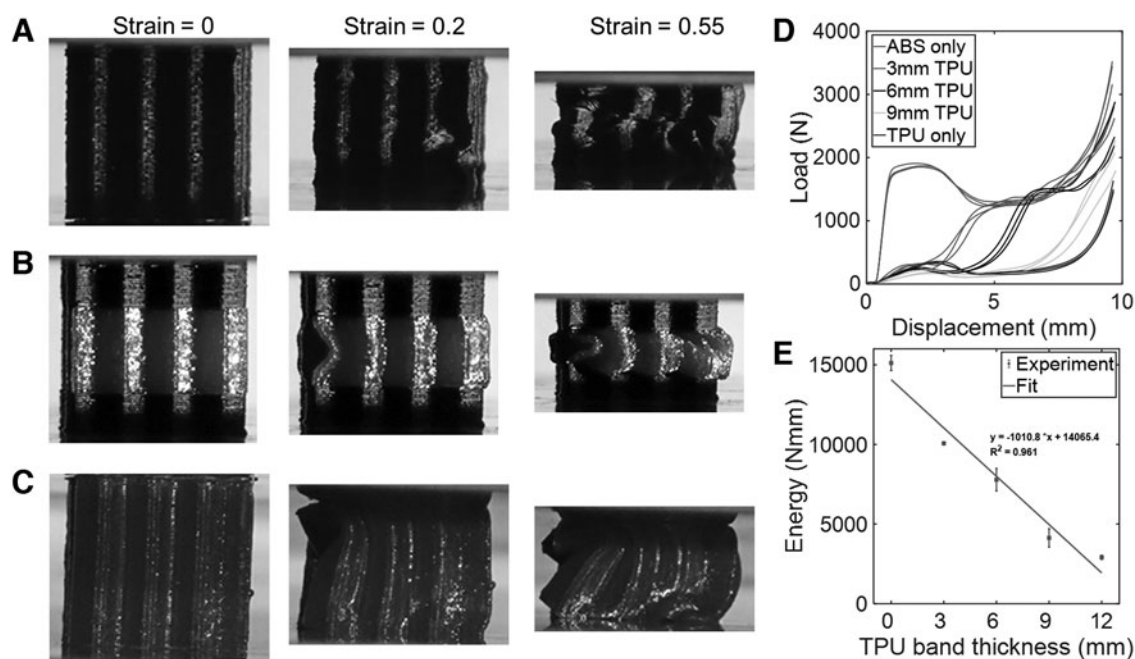


FIG. 8. Out-of-plane compression for hexagonal honeycombs for a (A) pure ABS honeycomb, (B) 50% ABS and 50% TPU multimaterial honeycomb, and (C) pure TPU honeycomb. (D) Load–displacement response of honeycombs with (E) energy absorption.

hexagonal lattice at different strains. The deformation starts from the unit cells below the diagonal (shown by red box) with the unit cells immediately under the diagonal deforming, followed by the failure of cells above the diagonal unit cells. Figure 10B shows the load–displacement curve for the in-

plane compression of the hexagonal honeycomb, and Figure 10C shows the energy absorption for different TPU band thicknesses. A similar trend of the compression energy absorption behavior is observed with the highest energy absorption of 1.96 ± 0.081 kN·mm for ABS-only samples and

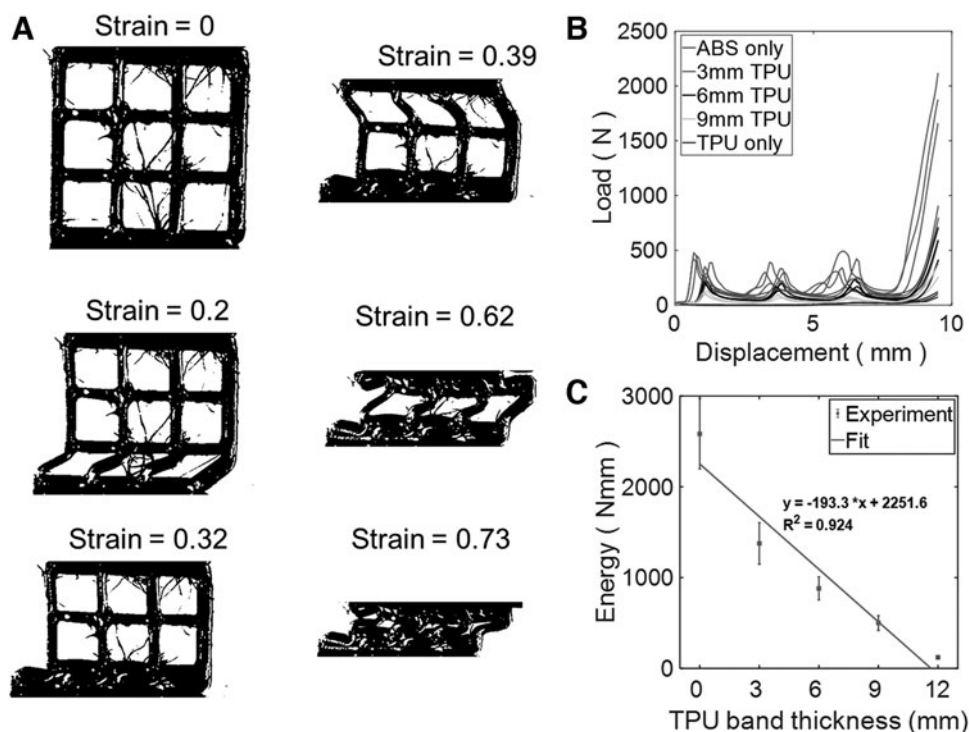


FIG. 9. In-plane compression of a *square* honeycomb sample demonstrating (A) failure stages at different strains when the TPU band is 6 mm, (B) load–displacement response of multimaterial samples, and (C) energy absorption.

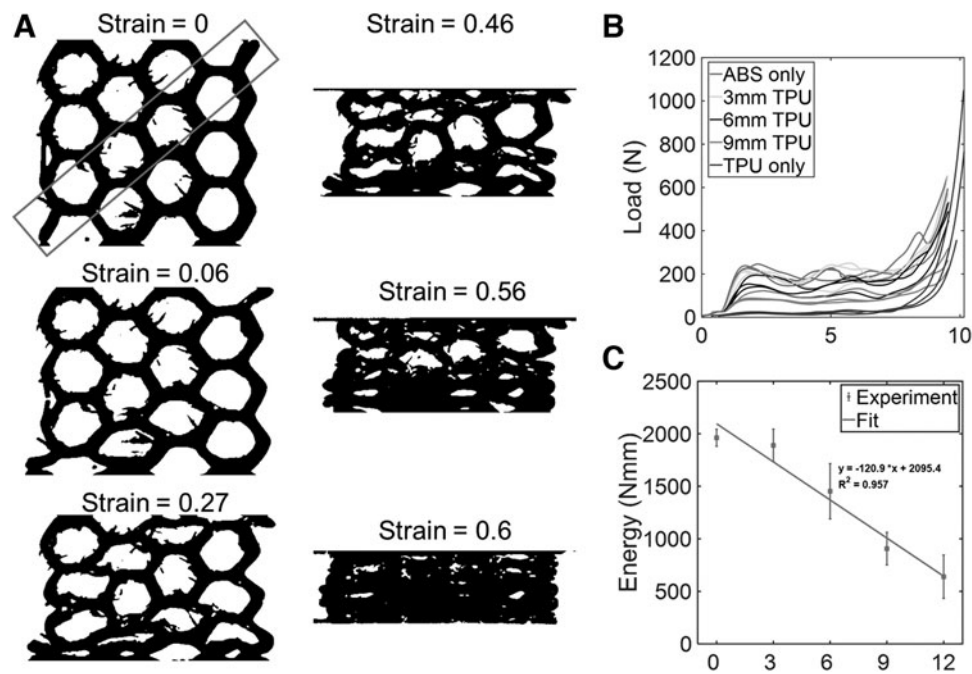


FIG. 10. In-plane compression of a *hexagonal* honeycomb sample demonstrating (A) failure stages at different strains when the TPU band is 6 mm; *grey rectangle* - initiation of deformation, (B) load-displacement response of multimaterial designs, and (C) energy absorption.

the lowest of 0.64 ± 0.2 kN·mm of energy absorption for TPU-only samples.

For all cases of mixed materials and TPU-only samples, the hexagon structure has a significantly higher energy absorption than the square design ($p \leq 0.05$), demonstrating the hexagonal structure as the preferred design for maximizing energy absorption for in-plane loading. The fitted equations in Figure 10C, when considered with Figure 10E equations, provide quantifications for tuning the hexagonal design appropriately for engineered systems for specified stiffness, strength, and elongation.

The energy absorption of all samples as the function of displacements up until 80% strain is provided for all structures in Figure 11. Each curve shows the mean of cumulative energy absorption for three samples up until each deformation point for each designed and tested structure. These plots demonstrate that trends persist across all displacement measurements that the higher proportion of ABS in a structure generally leads to higher energy absorption. However, in some cases such as out-of-plane testing, there is more nonlinearity present with larger differences in cumulative energy at high displacements, while at low displacements, all structures with the TPU band behave similarly. These results demonstrate measured energy responses of systems that engineers can reference during optimization to provide a tailored structure for ABS-to-TPU ratios based on the energy and displacement constraints of specific applications.

Discussion

New design spaces opened by 3D printing provide a chance to exploit tunable mechanical responses for engineering applications, which was investigated here for multimaterial

square and honeycomb lattices. Experiments demonstrated that multimaterial ABS and TPU combinations enabled different energy absorption magnitudes and had differing behaviors based on out-of-plane and in-plane orientations.

Initially, microscope images were collected of fabricated samples to determine printing quality and defects that may

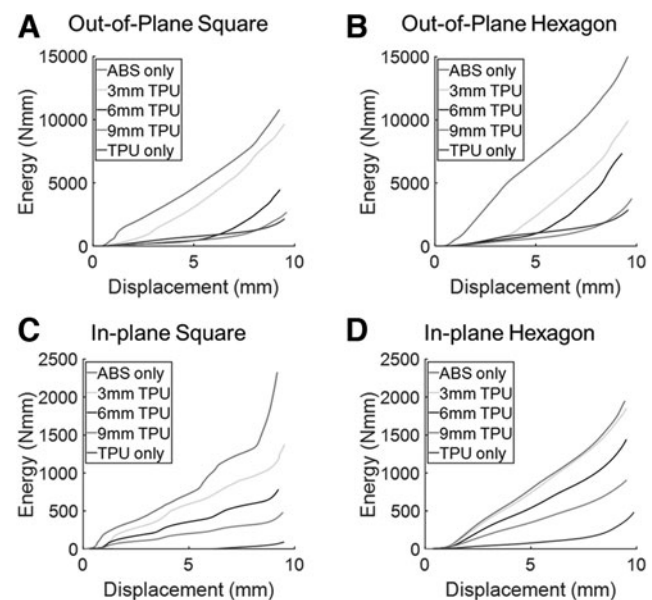


FIG. 11. Energy absorption as a function of compressive displacement for (A) out-of-plane *square*, (B) out-of-plane *hexagon*, (C) in-plane *square*, and (D) in-plane *hexagon* testing conditions.

affect mechanical response (Fig. 6). Printing defects in the form of microscopic holes were present in the ABS-printed portions of samples, while there was generally extra material due to stringing for TPU-printed portions, which is common in FDM-printed parts.⁴⁸ Visual inspection of the samples after the compression test showed that the ABS to TPU interface was more likely to delaminate compared with layers between the same materials. The likelihood to delaminate may be due to their thermal and viscosity differences and the order of material deposition between ABS and TPU.^{7,49} Since these defects were generally stochastic, they contribute to errors in the mechanical experiments and should be of concern to designers to consider uncertainty in performance due to manufacturing.

Mechanics were investigated initially for out-of-plane compression of the square and hexagonal honeycombs (Figs. 7 and 8). Comparisons between topologies were controlled by designing samples with a similar unit cell size, wall thickness, and overall dimension. The hexagonal honeycombs had significantly higher energy absorption for both out-of-plane and in-plane compressions. The energy absorption for the hexagonal specimen in out-of-plane compression was 66% higher than the square honeycomb for a TPU band of 6 mm size, which was the largest difference between the topologies. For in-plane compression, the energy absorption of ABS-only samples had no significance between the designs (Figs. 9 and 10), but again the hexagonal samples had a significantly higher energy absorption for all multimaterial and TPU designs. These findings suggest that hexagonal honeycombs are preferred over square honeycombs in all cases when maximum energy absorption is the primary design criteria.

A second design criterion that may be used is the displacement response of the structures under load. When ABS and TPU are combined, the materials have a trade-off between the displacement before failure and the stiffness. Generally, adding TPU to the system advances the peak load to a higher displacement, while adding ABS increases the magnitude of the peak load. Altering the load–displacement curve according to these rules provides a tunable energy absorption property by adjusting the relative proportion of ABS and TPU. These results demonstrate that the multimaterial approach is a comparable alternative to tailoring energy absorption by varying the relative density of unit cells throughout a structure.^{50,51} The initial deformation of TPU before ABS is similar to strategies for topology-graded structures where the lower density structure deformed before a high-density section.

An advantage of the multimaterial strategy is that all aspects of the structure are printable using the same feature size that enable two materials with vastly different properties to improve the overall structural responses. Graded structures, in contrast, are limited by a printer's resolution for altering relative density of a single material that could have a limited design space for volume-constrained applications.

These mechanical results inform applications for tuning contrasting materials to form high-efficiency structures that absorb energy, such as medical prosthetics and helmets.^{15,52} One example is tuning the materials based on a wearer's weight for prosthetics to achieve an optimized load–displacement response during repeated use. In addition, the use of flexible materials such as TPU is potentially beneficial for

reducing the impact of sudden loads,¹¹ while the use of ABS ensures that a high peak load is sustained. Some limitations in our approach are the need to consider the resizing of lattices and precise differences in measured values, such as material density, that could affect system optimization.

When resizing lattices by altering the number of unit cells, it is expected that the boundary conditions may play a role such that the energy absorption per volume differs slightly for larger systems of honeycombs with many more unit cells than the economical 3–4 unit cell systems studied here. Increasing the number of unit cells tends to result in convergent property values at larger numbers of unit cells.⁵³ The characteristic length scale for such studies is limited by the resolution of the printer, with a need for unit cells to be much larger than the thickness of each additive layer, but the number of unit cells to remain low so structures are printed in a reasonable amount of time. Further precision in analyzing results is also possible by considering the ~20% higher density of TPU compared with ABS.

However, in the context of this study, decreasing the material volume of TPU per sample to match the weight of ABS in samples would result in less relative contribution of TPU that further reinforces the much greater energy absorption response of ABS for a given displacement.

Conclusion

This work presents an experimental study of the compressive response of multimaterial honeycombs under uniaxial compression using ABS and TPU materials. The tuning of TPU band thickness contributes to an alteration in the compressive load–displacement response. The ABS imparts stiffness for samples, while TPU enables displacement before failure. By controlling the thickness of TPU, a structure's mechanical response is tunable for specified applications, such as adjusting how a structure displaces while absorbing energy for a leg prosthetic. Experiments demonstrated that the ABS-only samples have the highest energy absorption that decreases as the TPU band thickness increases for both square and hexagonal honeycombs. In-plane compression tests showed that structural failure occurs layer-by-layer for square honeycombs, while a progressive failure without any peaks occurs for hexagonal honeycombs. The differing failure modes provide contrasting behaviors that inform design decisions when using multimaterial structures.

Overall, these results demonstrate the merits of combining contrasting materials to form advantageous responses in structures that are suitable for diverse engineering applications.

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Authors' Contributions

N.R.K.: Conceptualization, methodology, data curation, and writing—original draft preparation. P.F.E.: Conceptualization, supervision, and writing—reviewing and editing.

Author Disclosure Statement

No competing financial interests exist.

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